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EXTENDED ABSTRACT

Multi-source cross-sensor harmonization via generative self-supervised learning

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1. Introduction and objectives

A significant fraction of satellite imagery used in meteorological data assimilation comes from constellations, i.e. ensembles of satellites coupled to produce a common product. An example of such constellation is the Global Precipitation Measurement (GPM), which includes different Passive Microwave (PMW) sensors, and a reference instrument, the GPM Microwave Imager (GMI). GMI is coupled with a radar aboard the same satellite, which allowed to develop statistical methods to estimate precipitation from PMW observations. Such methods include the GPROF algorithm [3], or recent deep learning approaches [5]. However GMI is the only sensor within the constellation that is paired with a radar; thus images from the other PMW sensors have to be transferred into the domain of GMI for the downstream statistical methods to properly apply. This task of going from the measurements of several sensors with varying characteristics (resolution, viewing angle, etc...) to a reference one is referred to as cross-sensor harmonization. It can itself be approached with deep learning, the most natural way being to train on coincidences (images at nearly the same time over the same area) between GMI and the other sensors.

In this preliminary work, we suggest that training only on coincidences with GMI is actually sub-optimal for generative reconstruction. In this setting, the data is limited to scarce coincidences between GMI and the other sensors in the constellation. Instead, we propose to train on images that are further apart in time and to use a more general training task, both of which increase the amount of training samples.

Contributions.

- We introduce the first generative model for cross-sensor harmonization of passive microwave images.
- We show that relaxing the spatio-temporal coincidence constraint and training using all satellites as targets benefit the results on the original problem despite making the training task more difficult.

2. Methods and data

We train a generative model based on Flow Matching [2]. The architecture is based on Motif [1], designed to fuse data from multiple geospatial sources that are misaligned in time and space. As the original architecture was designed for deterministic training, we adapted it for flow matching, notably by adding embedding layers for the flow matching step. We use the TC-PRIMED dataset [4], which

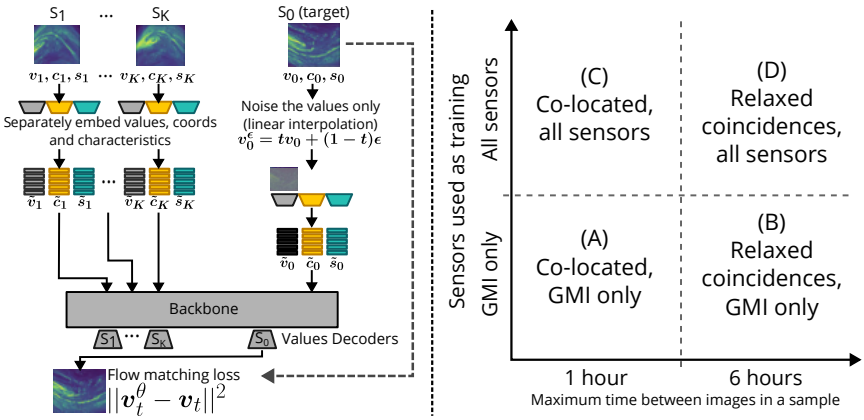

Figure 1. *Left: training pipeline: a set of sources $S_1, ...S_K$ are embedded into three embedding spaces (pixels, spatio-temporal coordinates, and source characteristics) and fed to the backbone. A target source S_0 is linearly interpolated with gaussian noise and then similarly embedded. The backbone predicts the reconstructed image, which is evaluated using the conditional flow matching loss (following [2]). Right: the four training configurations obtained by using or not each change.*

Table 1. *Comparison of CRPS values for different training configurations (lower is better). Brackets indicate bootstrapped 95% confidence intervals.*

Experiment	Coincidence strictness	Sensors used as targets	CRPS
(A)	Co-located images ($\Delta t = 1h$)	GMI only	7.59 [7.44,7.73]
(B)	Relaxed ($\Delta t = 6h$)	GMI only	6.54 [6.41,6.65]
(C)	Co-located images ($\Delta t = 1h$)	All 11 sensors	6.39 [6.28,6.51]
(D)	Relaxed ($\Delta t = 6h$)	All 11 sensors	**4.77** [4.69,4.85]

contains PMW images of over 3,000 tropical cyclones between 1987 and 2023, from 11 sensors aboard 11 satellites. Our experiment consists in training and evaluating the same architecture with identical training parameters (number of steps, learning rate, batch size, etc...), but in different configurations, whose details are displayed in Figure 1.

3. Preliminary results and/or expected outcomes

Table 1 shows the performance of the same architecture trained on each configuration, but evaluated exactly on the same samples. The evaluation samples are co-located images ($\Delta t = 1h$) in which the target is always GMI, since this is the task of interest. The metric used is the Continuous Ranked Probability Score (CRPS), with 10 members generated for each sample. Our results show that combining the two changes (relaxed coincidences and using all sensors as targets) significantly decreases the CRPS, showing an improved reconstruction of GPM images.

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Competing Interests. None

Data Availability Statement. All data used in this worked is available through the website and API of TC-PRIMED [4] at https://rammb-data.cira.colostate.edu/tcprimed/. The code can be accessed through its Zenodo repository

Ethical Standards. The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

Author Contributions. Conceptualization: C.D.; Data curation: C.D.; Funding acquisition: C.M.; Investigation: C.D.; Methodology: C.D.; Project administration: C.D. and C.M.; Resources: C.M.; Software: C.D.; Supervision: C.M.; Validation: C.D.; Visualization: C.D. and C.M.; Writing – original draft: C.D.; Writing – review and editing: C.M.

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